Why this matters

The edge of a wildfire is the most dangerous place for a firefighter. Robot dogs (four-legged robots) can climb steep, rocky, smoky ground where trucks get stuck and drones can't see.

But a robot near a fire must know where the fire **will** be, not just where it is now.

How we forecast the fire

Real fire history
~246,000 NASA satellite fire detections from 30 major wildfires worldwide

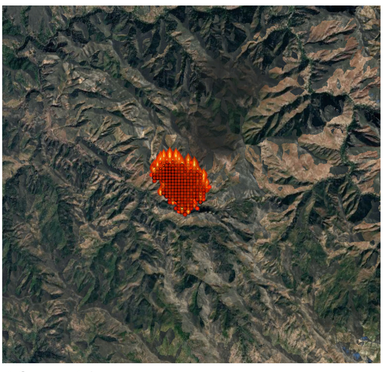
Conditions
weather (wind, humidity, temperature, rain), terrain (elevation, slope), land cover, and the fire's current shape

Machine learning (XGBoost)
27 inputs per 50 m cell. Trained on 24 fires, tested on 6 it never saw.

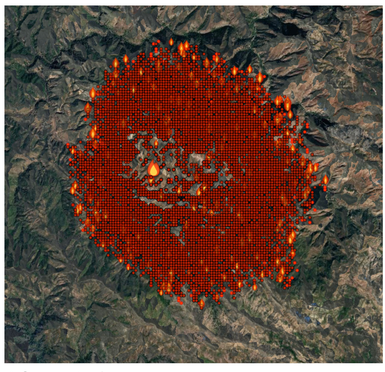
Physics (Rothermel spread equation)
fuel × wind × slope × dryness; the ML model adjusts the rate

Output
the chance each 50 m cell catches fire, every 15-minute step

The simulator



After ~4 hours



After ~1 day

Real 3D terrain near Big Sur, Monterey County, California (summer weather). Runs in a web browser on Google 3D map tiles.

Is the forecast any good?

Tested on 6 unseen fires (USA, Bolivia, Siberia, Greece). 91% of cases there are "fire spreads," so the baseline always guesses "spreads."

Ranking risky cells AUC-ROC, higher is better

Our model	**0.75**
Baseline	**0.50**

Probability error Brier score, lower is better

Our model	**0.076**
Baseline	**0.083**

Clearly better than chance at ranking risk, with more accurate probabilities than the baseline. It still leans toward "spreads," which we are working on.